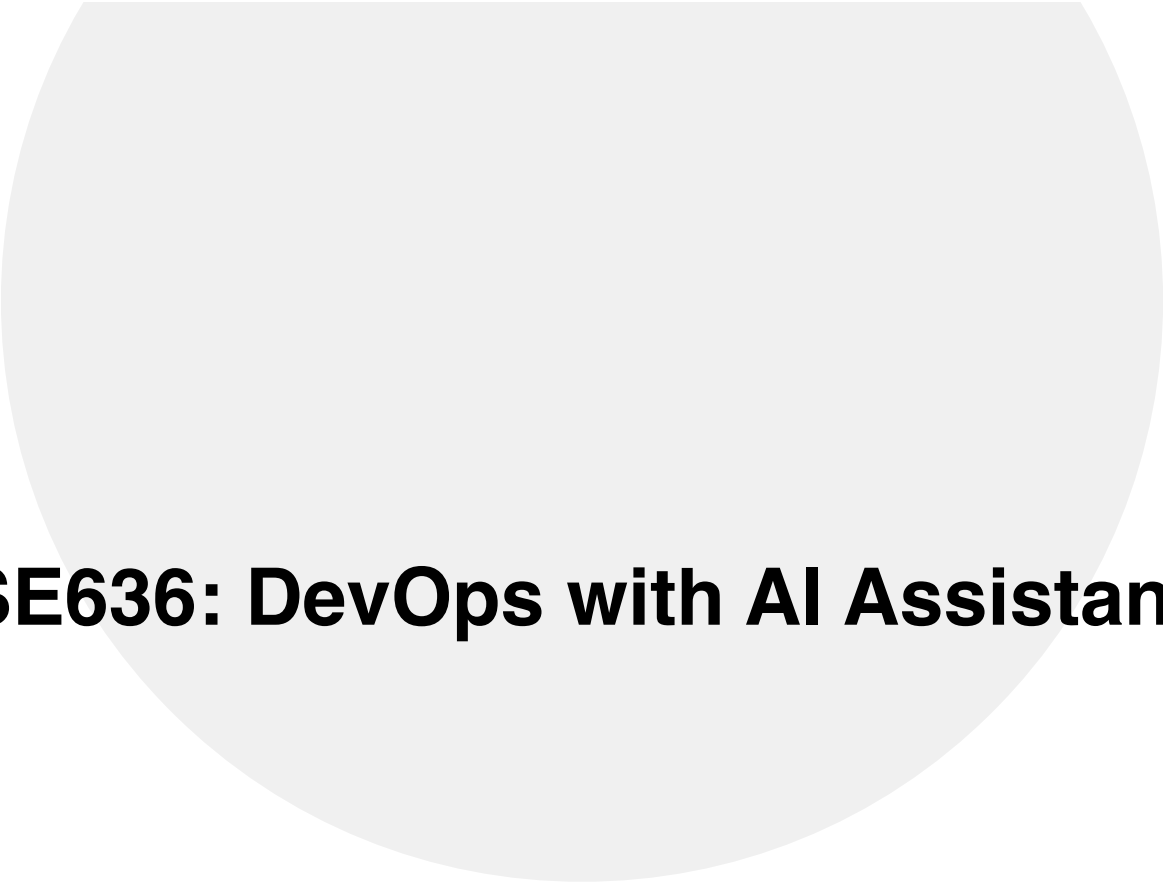
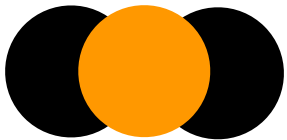
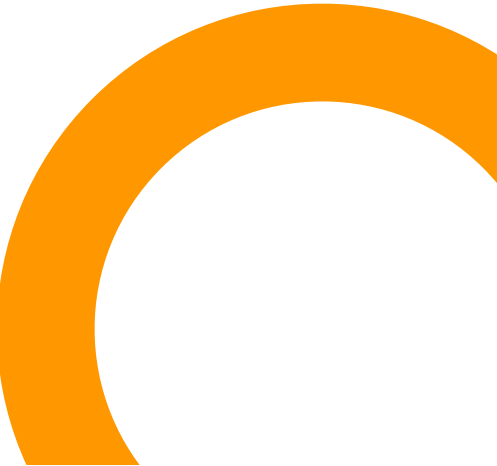




CSE636: DevOps with AI Assistance

Fall, 2025

Qingsong Zhang, Ph. D.



Meet Your Professor

Academic Background

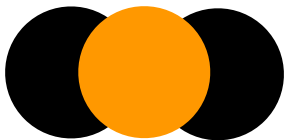
Ph. D. In Engineering, McGill University, Montreal, Canada
Mathematical model, Signal automation, Coding, Control
Language: Basic, Fortran, Cobol, Assembly, Pascal, C, C++, Java, SQL, MATLAB
System: MainFrame, 8086/80286/80386, PC, IBM, MAC, HPUX, SUN
Network: Novell, Token Ring, Ethernet, LAN, WAN, SD-WAN, PBX

Professional Expertise

Hewlett Packard, IT Specialist —> Software Release Manager
General Electric, Software Release Manager and Installer
Affymetrix, Software Development Manager
GE Digital, DevOps Manager, Cloud Infrastructure Staff
Walmart eCommerce, DevOps Director, Infra/QA/Support
PayPal, TPX (Technology, Platform and Technology) Director
Michaels, VP of Technology, Infra, Operation (SRE) and Big Data
PayPal, Principal Architect, SRE, Observability, Telemetry, AI Agent, MCP

Teaching

Northeast Polytechnic University (SFBU), 15+ years, Unix, Network,
Oracle, QA, DevOps (Agile)
California Science and Technology University (CSTU), 2+ years,
ChatGPT, DevOps+AI



Course Information

01 Lecturer: Qingsong Zhang
TA: Van Lam

02 E-mail: qingsong.zhang@cstu.edu
van.lam@cstu.edu

03 Time: Wed 7:30–9:00pm (Online),
Sat 9:00–10:30am (Onsite)

04 Length: 23 Hours | Credit: 1.5
Units

05 Office Hours: By Appointment
Review Date: 11/4/2025





Grading Components

- Mid-term Exam: 15%
- Project: 20%
- Case Discussions: 30%
- Final Exam: 20%
- Class Participation: 15%

****Total: 100%****

Grading Scale

98–100%: A+
93–97%: A
90–92%: A-
88–89%: B+
83–87%: B
80–82%: B-
78–79%: C+
73–77%: C
70–72%: C-
68–69%: D+
63–67%: D
60–62%: D-
Below 60%: F

Course Summary: DevOps with AI

Course Overview

This 1.5-unit course explores how AI and ML can be integrated into DevOps practices to automate, optimize, and enhance software development and operations.

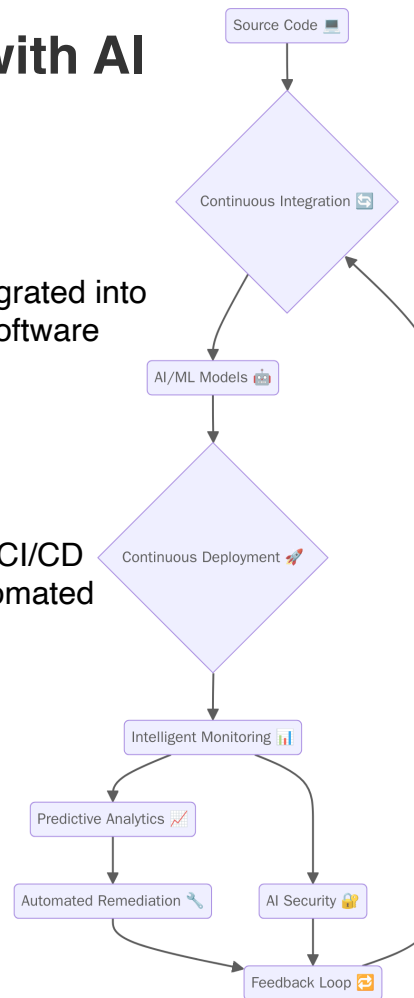
Practical Experience

Students will gain hands-on experience with AI-assisted CI/CD pipelines, predictive analytics, intelligent monitoring, automated remediation, and security.

Learning Outcomes

By the end of the course, students will:

- 1) Understand AI in DevOps,
- 2) Implement AI-driven automation,
- 3) Apply predictive analytics,
- 4) Use AI for monitoring and response,
- 5) Evaluate AI frameworks at scale.





Session Overview: What You'll Learn

-  **01** 1. Evolution of DevOps and need for AI
-  **02** 2. Key DevOps challenges today
-  **03** 3. AI vs traditional automation
-  **04** 4. AIOps and practical use cases
-  **05** 5. Industry trends and case studies





What is DevOps?

Origins and Philosophy

DevOps emerged to bridge the gap between software development and IT operations, emphasizing collaboration, automation, and continuous delivery.

Key Practices

Core practices include CI/CD, infrastructure as code, monitoring, and rapid feedback loops—all aimed at streamlining software delivery.

Why It Matters

DevOps accelerates innovation and reduces time-to-market, making it essential in today's fast-paced tech world.



What is DevOps?

DevOps

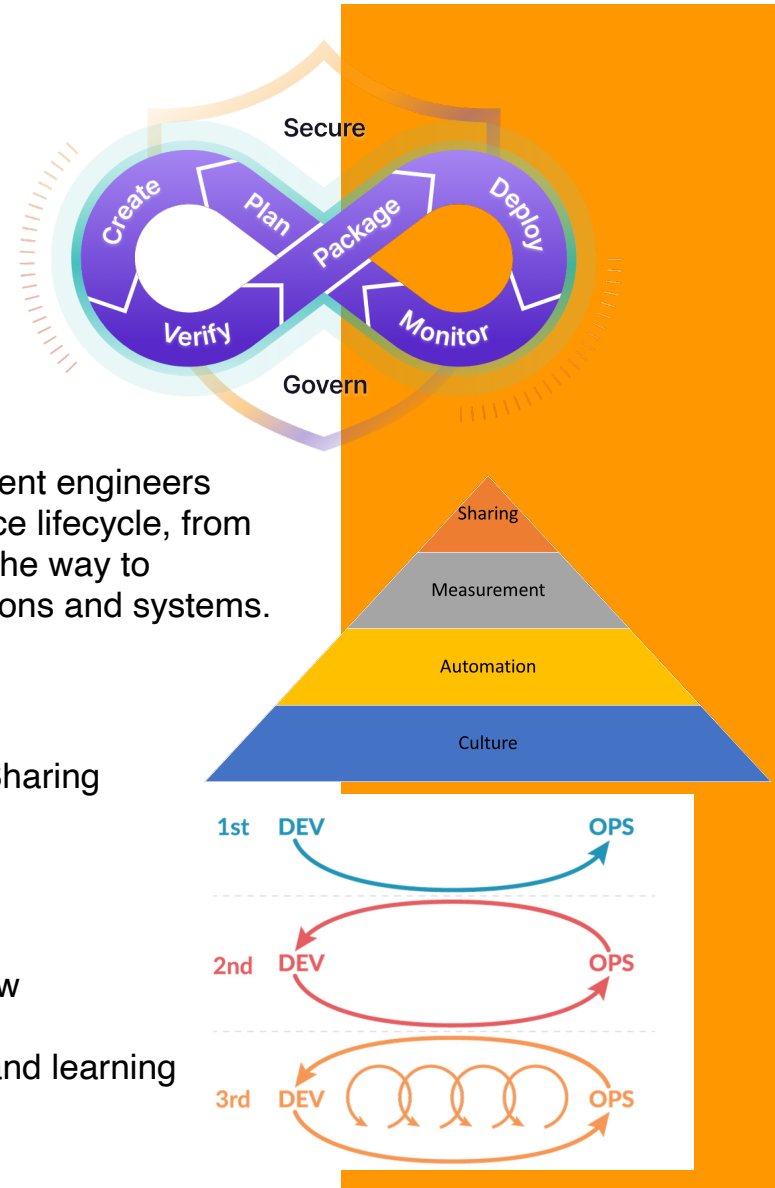
The practice of operations and development engineers working together through the entire service lifecycle, from the design and development process all the way to production support including the applications and systems.

CAMS Model

Culture, Automation, Measurement and Sharing

Principles: The Three Ways

Systems thinking and the principles of flow
Amplifying feedback loops
A culture of continuous experimentation and learning





Rise of AI in Software Engineering

AI Enters DevOps

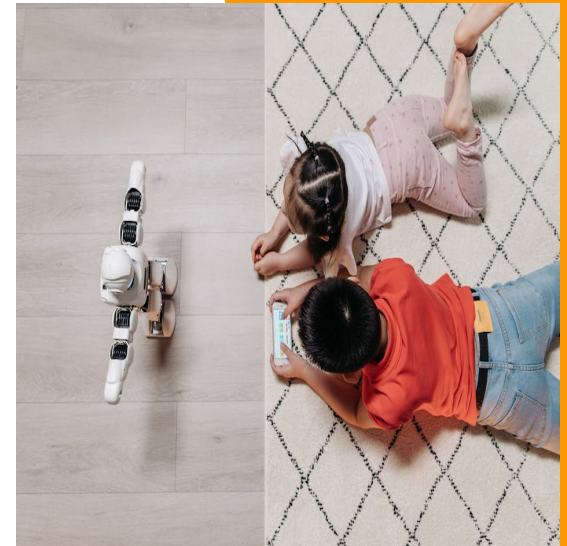
Artificial Intelligence has gradually moved into the DevOps lifecycle, particularly in testing, monitoring, and issue resolution.

Drivers of Adoption

Rising system complexity, data volume, and expectations for uptime have made AI integration increasingly attractive.

Impact on Engineering

AI tools help reduce manual tasks, improve decision-making, and detect issues earlier in the development process.





Why AI is a Game-Changer for DevOps

From Automation to Intelligence

AI elevates DevOps beyond rule-based automation by enabling systems to learn and adapt. This allows for smarter decision-making and proactive incident management.

Enhanced Speed and Reliability

AI reduces human errors and speeds up deployment cycles by predicting issues before they happen. It brings consistency and efficiency across environments.

Scalability with Insight

AI tools analyze massive amounts of data in real-time, helping teams scale DevOps practices without losing visibility or control.





Why AI + DevOps?

Complexity at Scale

Modern systems generate enormous amounts of telemetry data, making manual monitoring and response increasingly impractical.

Speed Expectations

Organizations demand faster deployments and minimal downtime, which requires intelligent automation to maintain reliability.

Adaptive Learning

AI can analyze historical patterns to predict future issues, proactively helping teams before problems impact users.



AI Across the DevOps Lifecycle

From Automation to Intelligence

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AI Across the DevOps Lifecycle

Code & Build

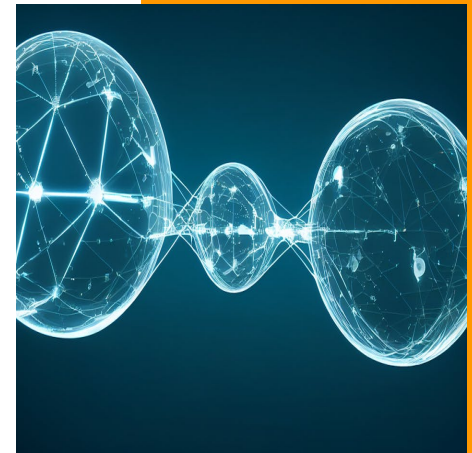
AI enhances code reviews, suggests improvements, and predicts bugs early in the pipeline, improving code quality before deployment.

Test & Deploy

Machine learning models optimize test case selection and deployment strategies, minimizing downtime and failures.

Monitor & Operate

Real-time anomaly detection and predictive analytics enable proactive issue resolution and smart alerting systems.





Benefits of AI in DevOps



Faster Deployments

AI can automate repetitive processes and decision points, helping teams push changes to production more frequently and safely.



Improved Stability

With real-time monitoring and learning-based alerting, systems can react to problems before they affect users.



Smarter Resource Allocation

AI-driven insights guide teams on how to optimize cloud usage, manage pipelines, and allocate developer attention effectively.



```


Home



Home Events



Multiple Column Menu



Tall Button Header



Image Logo



Tall Logo



Variable Width Slider



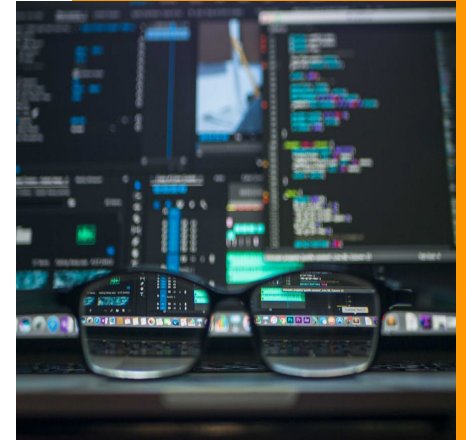
Testimonial


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Challenges of Integrating AI in DevOps

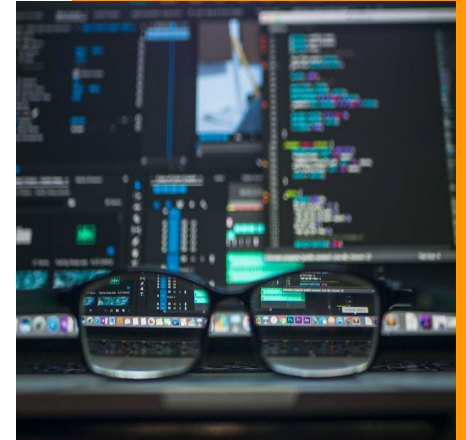
- **Data Quality and Availability** Effective AI models need high-quality, labeled data, which may be hard to obtain or organize in DevOps environments.
- **Model Transparency** DevOps teams may face trust issues due to the black-box nature of some AI algorithms, especially in critical systems.
- **Skill Gap and Tooling** Bringing AI into DevOps workflows requires new skillsets, proper integration tools, and cultural readiness across teams.





Challenges of Integrating AI in DevOps

- **Data Quality and Availability** Effective AI models need high-quality, labeled data, which may be hard to obtain or organize in DevOps environments.
- **Model Transparency** DevOps teams may face trust issues due to the black-box nature of some AI algorithms, especially in critical systems.
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Scalability Challenges in DevOps

Growing Infrastructure

Modern architectures like microservices and containers increase the complexity of managing environments at scale.

Human Limitation

Manual oversight struggles to keep up with real-time monitoring and dynamic scaling requirements.

AI to the Rescue

AI assists by automating responses and detecting patterns across massive datasets.





Monitoring and Incident Response

Traditional Monitoring

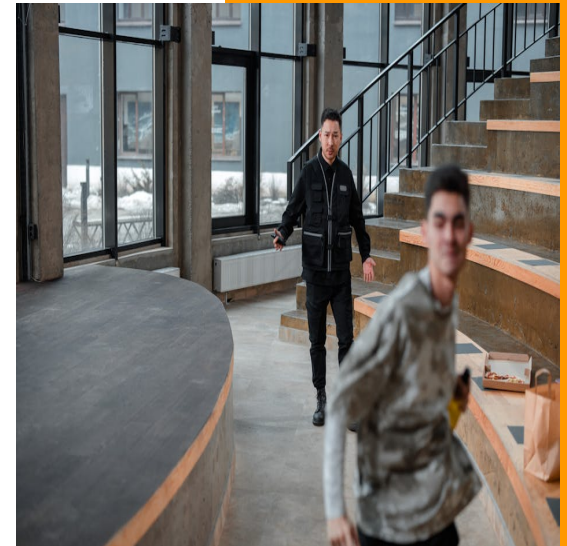
Relies heavily on pre-defined rules and manual analysis, which often leads to alert fatigue.

Challenges in Incident Triage

Teams face delays in identifying root causes and prioritizing the right issues to fix.

AI-Augmented Solutions

Machine learning models detect anomalies early, group related alerts, and recommend fixes.





Toolchain Complexity in DevOps

Too Many Tools

Modern DevOps stacks include a mix of CI/CD, monitoring, infrastructure, and collaboration tools.

Integration Fatigue

Connecting and managing toolchains requires effort and introduces potential failure points.

AI Integration Advantage

AI can unify and make sense of data from across the toolchain, enhancing decision-making.





Traditional Automation in DevOps

Scripted Workflows

Classic DevOps automation relies on scripts and tools configured for specific tasks like deployment or testing.

Benefits

It increases speed and reduces manual error but lacks adaptability.

Limitations

Static rules can't handle unexpected events, making them reactive rather than proactive.





AI-Driven DevOps

Proactive Intelligence

AI can analyze patterns, predict incidents, and recommend preventive actions before users are impacted.

Smart Decision-Making

Machine learning models help prioritize alerts, assign tasks, and optimize workflows in real-time.

Key Difference

Unlike static automation, AI evolves and adapts over time, improving efficiency and reliability.



Understanding AIOps

Definition

AIOps stands for Artificial Intelligence for IT Operations, combining big data and ML to automate and enhance operations.

Core Functions

AIOps platforms ingest logs, metrics, and events to detect anomalies, correlate incidents, and trigger responses.

Value Proposition

It reduces mean time to resolution (MTTR) and improves service reliability.





AI Use Case: Anomaly Detection

What It Does

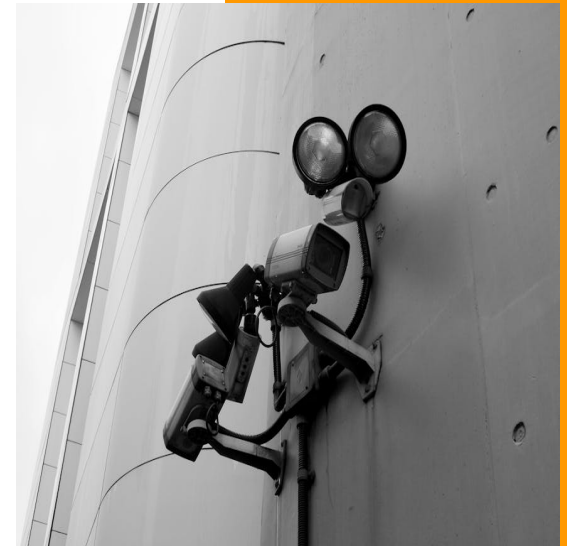
Automatically identifies deviations from normal behavior in logs, metrics, and performance data.

Why It Matters

Early detection helps prevent outages and performance degradation.

Student Perspective

Visualize how your future monitoring tools might detect a server issue before it affects users.





AI Use Case: Resource Optimization

Dynamic Scaling

AI helps adjust compute and storage resources based on real-time usage patterns.

Cost Efficiency

Smart resource allocation avoids over-provisioning, cutting unnecessary costs.

Example

A system detects low traffic periods and automatically scales down infrastructure without service impact.





AI Use Case: Incident Triage

Alert Overload

DevOps teams face too many alerts, many of which are false positives or low priority.

AI Filtering

AI clusters alerts by root cause and urgency, allowing teams to focus on real issues first.

Student Takeaway

Imagine being able to focus only on problems that truly matter—AI makes that possible in high-noise environments.





Case Study: Netflix AIOps

The Challenge

Managing a vast microservices ecosystem and minimizing user-facing issues in real time.

The Solution

Netflix uses AI to analyze metrics and logs, detect anomalies, and route incidents automatically.

The Outcome

Improved uptime, faster resolution, and better user experience at scale.





AI Tooling Landscape (2025)

Monitoring & Observability

Tools like Dynatrace, DataDog, and New Relic are integrating more AI features for alert reduction and root-cause analysis.

CI/CD & Deployment

GitHub Copilot, Harness, and others provide AI-driven code suggestions and deployment strategies.

Trends to Watch

Consolidation of platforms, low-code AIOps, and open-source innovation are shaping the future of AI in DevOps.



What is Generative AI?

Creating, Not Just Analyzing

Generative AI refers to models that can produce new content—like code, text, or images—based on training data and patterns.

Why It Matters in DevOps

DevOps teams can use generative AI to generate config files, deployment scripts, or documentation, saving time and improving quality.

Common Examples

Tools like ChatGPT and GitHub Copilot showcase how generative AI can support day-to-day developer and ops work.

```
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LLMs: Large Language Models in DevOps

What is an LLM?

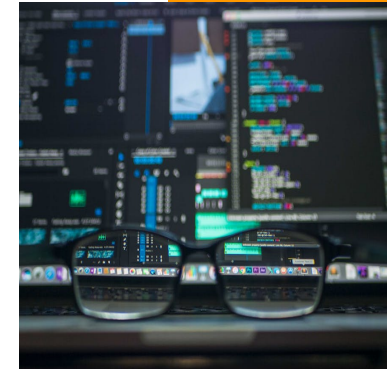
LLMs are powerful AI systems trained on massive text datasets that understand and generate human-like language.

Use in DevOps

They assist in writing shell scripts, analyzing logs, generating YAML files, and answering complex DevOps queries quickly.

No Theory, Just Use

Without diving into neural network theory, LLMs work like smart assistants—ready to solve real-world DevOps problems.



Prompt Engineering Basics

What is a Prompt?

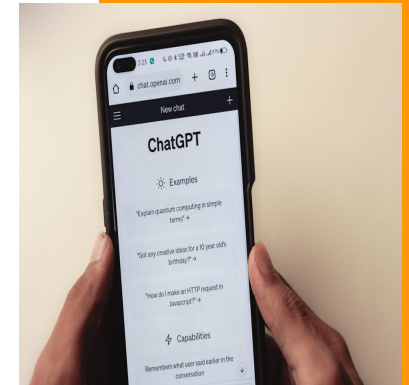
A prompt is the input you give an AI model to get the desired output, like 'Write a bash script for CPU monitoring'.

Crafting Effective Prompts

Be specific, include context, and define format to get accurate responses from LLMs.

DevOps Use Cases

From CI/CD pipelines to infrastructure provisioning, prompts can drive AI to generate useful artifacts instantly.



Traditional Automation vs AI-Driven

■ Rule-Based vs Intelligent

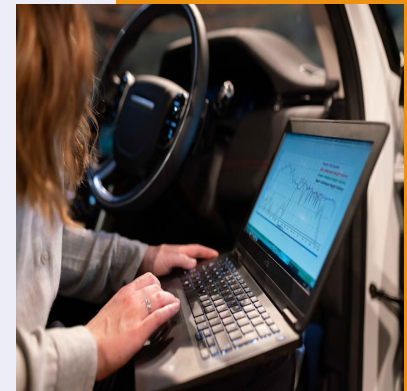
Traditional DevOps automation relies on predefined rules. AI can adapt and learn from new data, making decisions without explicit instructions.

■ Static Scripts vs Dynamic Generation

AI enables dynamic creation of scripts and configurations tailored to real-time inputs and context.

■ Error Handling & Recovery

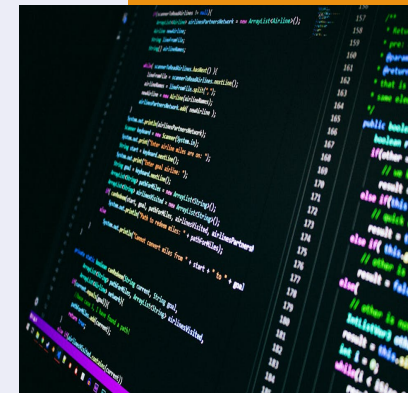
AI systems detect anomalies and recommend or execute recovery actions autonomously, reducing downtime and human effort.





GitHub Copilot

- **AI-Powered Code Assistant** GitHub Copilot suggests whole lines or blocks of code as you type, trained on public codebases, including shell and infra scripts.
- **DevOps Use Cases** Copilot helps with writing CI/CD scripts, Dockerfiles, and configuration management in YAML or JSON.
- **Pros & Watch-outs** Accelerates development, but results should be reviewed for accuracy and security, especially in ops code.





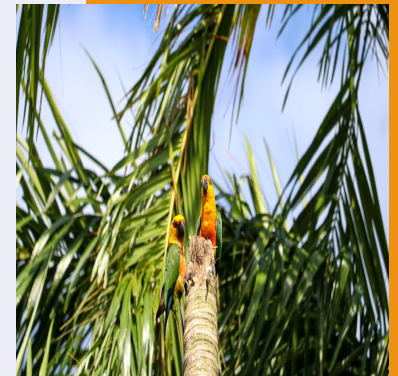
Amazon CodeWhisperer

- **Developer-Focused AI Suggestions**

Amazon CodeWhisperer is an AI coding companion that provides real-time code recommendations tailored to AWS services.
- **DevOps-Relevant Features**

Supports languages like Python, Bash, and JavaScript—useful for scripting automation and managing infrastructure.
- **Enterprise Ready**

Offers integrations with AWS IAM, logging, and enterprise-grade security for DevOps workflows.



OpenAI API

Flexible LLM Capabilities

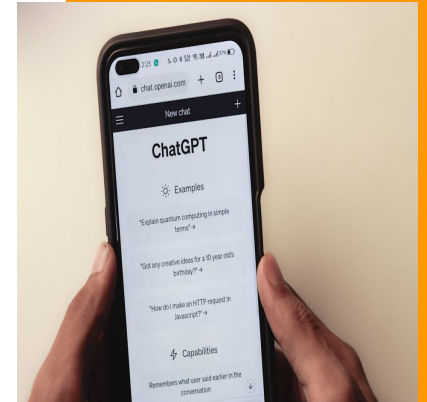
OpenAI's GPT-based API allows DevOps teams to integrate intelligent chat, document generation, and scripting assistance into tools.

DevOps Integration

Useful for log summarization, YAML generation, and explaining error messages in plain English.

Use Responsibly

While powerful, API usage should include safety guards, token limits, and audit trails in DevOps environments.





Ansible + AI Integrations



Smart Configuration Management

Ansible modules can be extended with AI to predict deployment issues, validate playbooks, and generate templated configs.



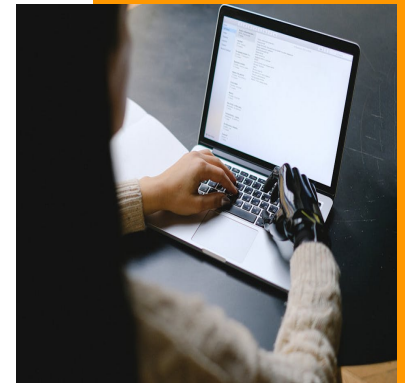
Intelligent Playbook Reviews

AI can audit or suggest improvements in playbooks based on past runbooks and common patterns.



Faster Remediation

AI tools can read Ansible logs and recommend fixes for failed runs automatically.





Kubernetes with AI Monitoring



Complex System, Smarter Monitoring

AI-enhanced monitoring solutions can detect issues in Kubernetes clusters before they escalate by analyzing logs, metrics, and trends.



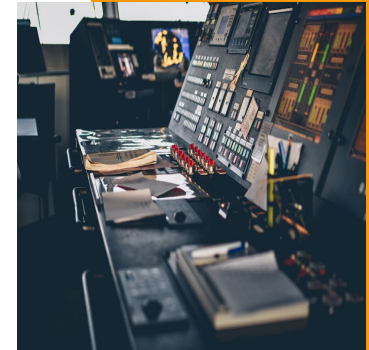
Root Cause Analysis

Machine learning models help trace performance issues to their source faster than traditional dashboards.



Better Scaling Decisions

Predictive scaling uses AI to allocate or remove resources proactively based on usage patterns.





Summary: Popular AI Tools for DevOps



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GitHub Copilot – Code generation and suggestions



Amazon CodeWhisperer – AWS-optimized coding assistant



OpenAI API – Flexible LLM integration for custom workflows



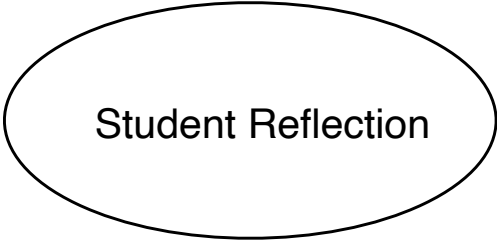
Ansible + AI – Smarter playbook validation and automation



Kubernetes AI Tools – Predictive monitoring and scaling



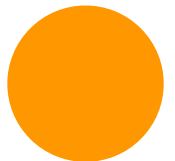
Looking Ahead: Your Role in AI + DevOps



Student Reflection

How ready is your DevOps
pipeline for AI?

As future engineers, you will shape how AI transforms DevOps. Think critically about automation, ethics, and innovation. Ask yourself how you would approach integrating AI into your own workflows.



Hands-On Project: VM Health Checker



What You'll Build

A bash script that checks key system parameters—CPU usage, memory, and disk space—and reports health status.

Interactive Mode with 'explain'

The script accepts an 'explain' argument to provide a detailed, user-friendly summary of the VM's condition.

DevOps Impact

Automating health checks helps teams monitor system integrity and proactively detect issues before outages occur.

Step 1: GitHub Repository Setup



Create Your Repo

Start by creating a new GitHub repository named ``vm-health-check``. Initialize it with a README and a ``.gitignore`` for shell scripts.

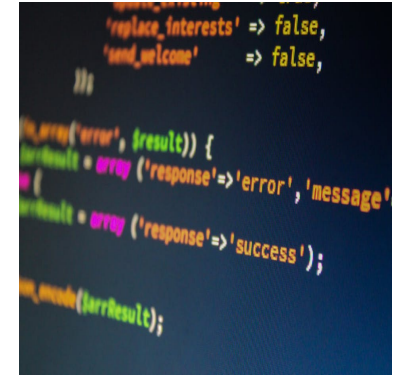
Folder Structure

Organize your files with a clear folder structure, e.g., ``.scripts/health_check.sh`` and ``.docs/usage.md``.

Version Control

Commit changes regularly with meaningful messages to track script improvements.

Step 2: Script Structure Overview



```
    'replace_interests' => false,
    'send_welcome' => false,
  })
  try {
    result = array ('response'=>'error', 'message'
  ) (
    result = array ('response'=>'success');
  }
  console.log(result);
```

■ Shell Script Entry Point

Create a `health_check.sh` file with a `main()` function. Use modular functions for CPU, memory, and disk checks.

■ Argument Parsing

Support `--explain` using a `case` block or `getopts` to trigger a verbose summary mode.

■ Execution Flow

Check parameters > run health checks > generate output > optionally display explanations.

Step 3: CPU Health Check



CPU Usage Monitoring

Use ``top``, ``mpstat``, or ``/proc/stat`` to gather CPU usage. For example: ``top -bn1 | grep 'Cpu(s)'`.

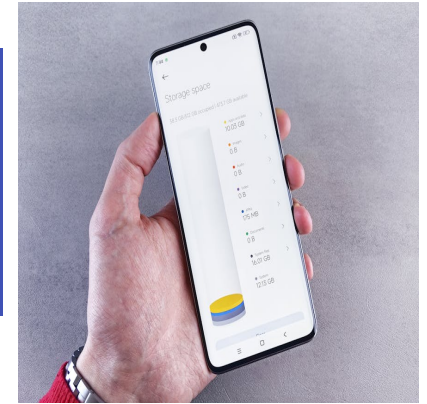
Set Thresholds

Compare current usage against a defined threshold (e.g., 85%) to determine health status.

Script Snippet

Example: ``if (( $(echo "$cpu > 85" | bc -l) )); then echo "High CPU usage"; fi``

Step 4: Memory Health Check



Capture Memory Usage

Use ``free -m`` to retrieve memory statistics and calculate usage:
``used/total` * 100.`

Set Safe Threshold

A usage over 90% might indicate memory pressure. Flag as 'Warning' or 'Critical' accordingly.

Script Snippet

Example: ``mem_usage=$(free | awk '/Mem/ { printf("%.2f", $3/$2 * 100.0) }')``

Step 5: Disk Space Check



Gather Disk Usage

Use ``df -h /`` or ``df --output=pcent / | tail -1`` to get the root filesystem's usage percentage.

Threshold Alerts

Mark disks as unhealthy if usage exceeds, say, 80%. Show warnings in red or bold terminal output.

Script Snippet

```
disk_usage=$(df / | awk 'END { print $5 }' | sed 's/%//')
```

Step 6: Combine All Health Metrics



■ Unified Health Logic

Run all three checks—CPU, memory, and disk—and aggregate their results into a single status report.

■ Visual Output

Use colors and symbols (✓ / ⚠ / ✗) to show system health per category.

■ Return Codes

Return specific exit codes (0: OK, 1: Warning, 2: Critical) to allow integration with cron or monitoring systems.

Step 7: Implement --explain Argument



Extended Output Mode

Add logic to detect `--explain` and display verbose descriptions for each metric's status.

Human-Friendly Language

Translate raw values into plain language. For example, 'CPU usage is above threshold. This may affect performance.'

Optional JSON or Markdown

Support output in structured formats for integration with monitoring tools or dashboards.

Step 8: Add Logging & Formatting



■ Log Important Events

Use ``logger`` or redirect output to a log file for tracking script execution and results.

■ Terminal Aesthetics

Use ANSI color codes for red (critical), yellow (warning), and green (healthy) statuses to improve readability.

■ Error Handling

Include fallback logic for missing tools or malformed inputs to make the script robust.

Step 9: Test Across VM Types



Run on Different Environments

Test the script on small, medium, and large VMs to ensure thresholds are realistic across contexts.

Validate --explain Output

Check if the extended output works as expected across varied loads and edge cases.

Automate Testing

Use cron jobs or CI pipelines to run and verify health checks periodically.

Step 10: Script Usage Examples



Basic Execution

Run `./health_check.sh` to print the current health status of the VM to the terminal.

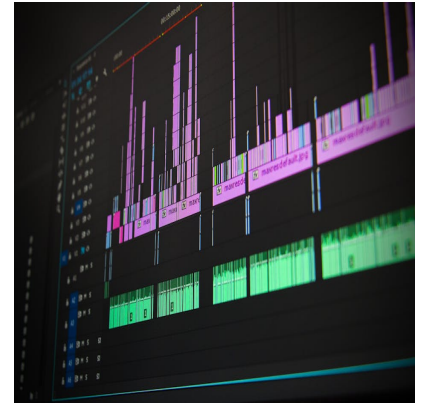
Verbose Mode

Run `./health_check.sh --explain` to get detailed output explaining each metric and its implications.

Automated Monitoring

Integrate into cron with ``*/15 * * * * /path/to/health_check.sh >> /var/log/vm_health.log``

Step 11: Automate Health Monitoring



■ Using Cron Jobs

Schedule the script to run periodically using ``crontab -e`` with your preferred timing interval.

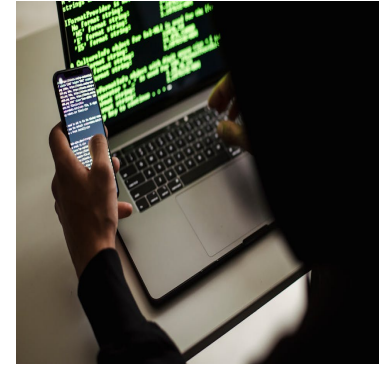
■ Email Alerts

Use ``mail`` or ``sendmail`` to notify ops teams of unhealthy statuses directly from the script.

■ Integrate with Dashboards

Log results in a format consumable by Prometheus/Grafana or custom monitoring UIs.

Step 12: Walkthrough Demo



Demo Objectives

Show how the script runs, outputs statuses, and responds to `--explain`.

Recording or Live Run

Use screen recording or live terminal share to walk through setup, usage, and output.

Validation

Confirm script behavior under high CPU, low memory, or full disk scenarios.